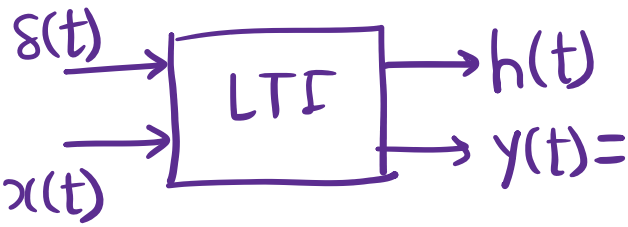
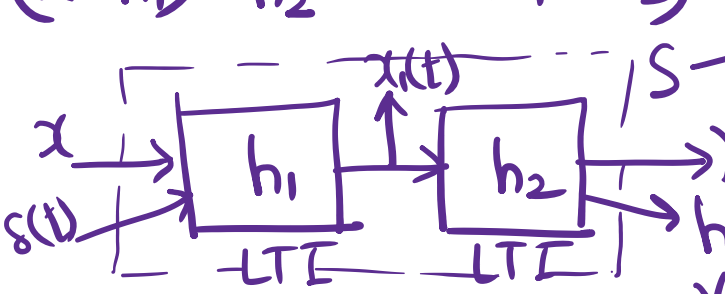
LTI System :



$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$(x * h_1) * h_2 = x * (h_1 * h_2)$ ∴ Proved.



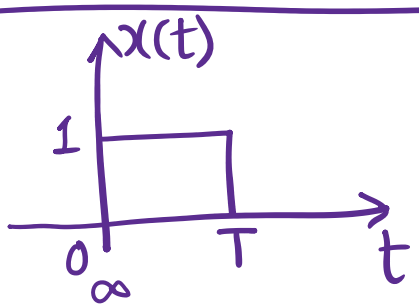
$y(t) = (x(t) * h_1(t)) * h_2(t)$ — ①

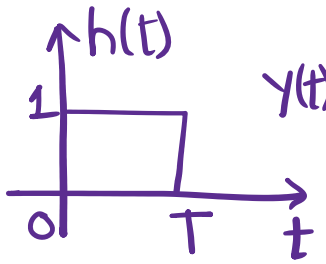
$h_1(t) * h_2(t) = h(t)$

$y(t) = x(t) * h(t)$

$y(t) = x(t) * (h_1(t) * h_2(t))$ — ②

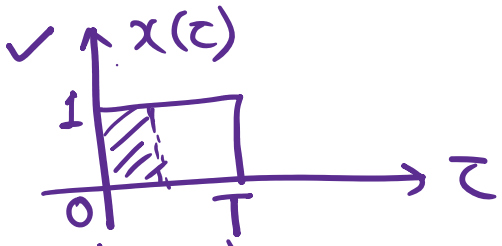
Ex:



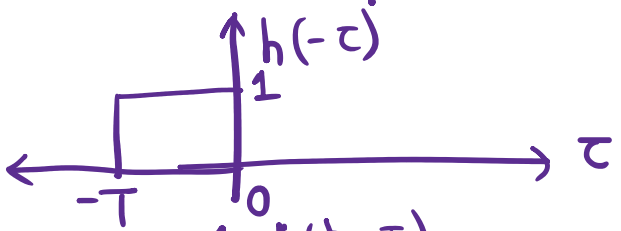


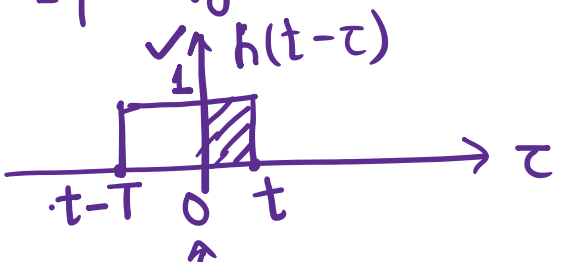
$y(t) = x(t) * h(t) = ?$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$



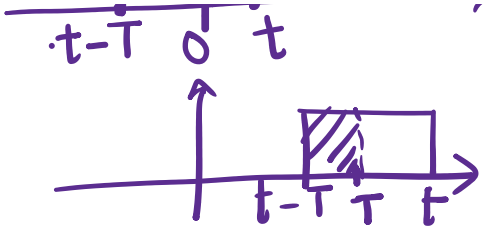
$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$





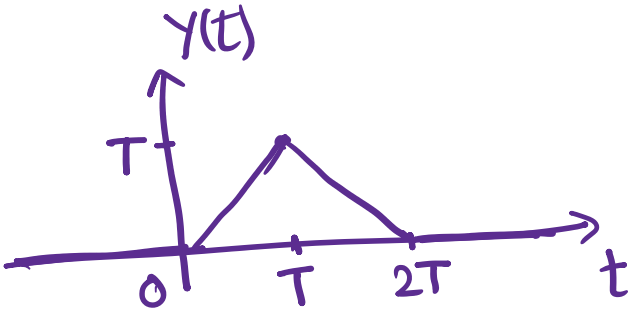
$0 \le t \le T$ $\quad y(t) = t$

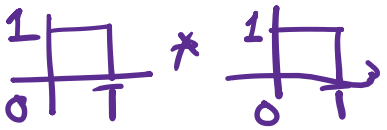
$t > T$ $\quad y(t) = 2T - t$



$\left.\begin{array}{l} t \geq T \\ \& \ t \leq 2T \end{array}\right\}$ $\quad y(t) = 2T - t$

$\& \ y(t) = \begin{cases} 0 & , t < 0 \ \checkmark \\ 0 & , t \geq 2T \end{cases}$





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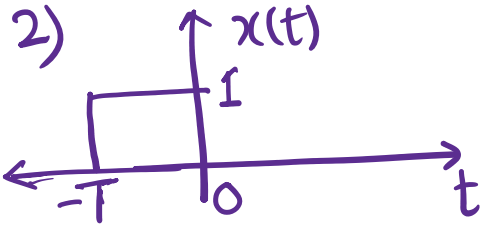
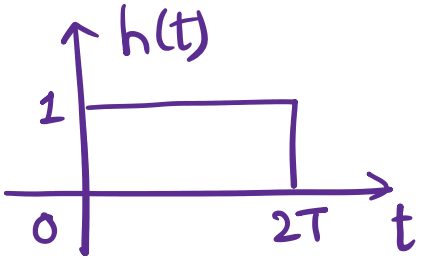
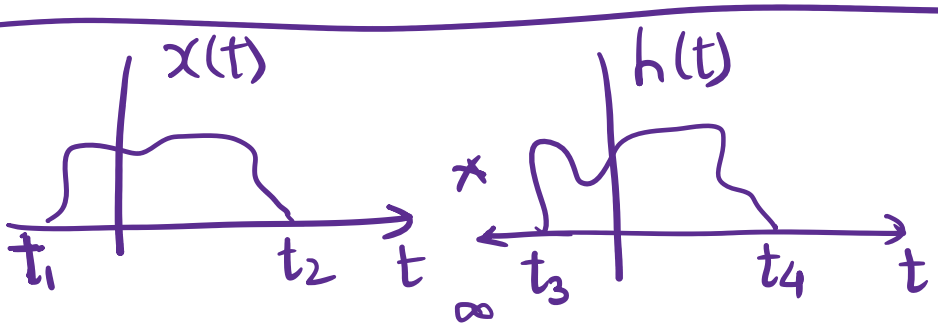
2)



$*$



$y(t) = (x * h)(t)$
$= ?$



$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) \, d\tau$$

☆ $y(t)$ can be non-zero only betn $[t_1 + t_3, t_2 + t_4]$ } Support of $y(t)$

Pf:

✓. $x(\tau)$: $\quad t_1 \leq \tau \leq t_2$

✓. $\underbrace{h(t-\tau)}_{P}$: $\quad t_3 \leq t - \tau \leq t_4$

$$t_1 + t_3 \leq t \leq t_2 + t_4$$

$y(t) = x(t) * h(t)$

$\downarrow$

$\boxed{h(t)}$

$$\boxed{h(t)}$$
LTI

1) Memoryless?
2) Causal/NC?

1) Memoryless/With memory (LTI)

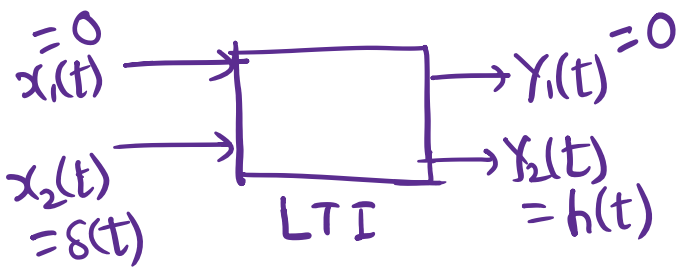
$y(t) = f(x(t))$ $\qquad x^2(t), |x(t)|, \ldots kx(t) \ldots$

$y(t) = \alpha x(t) \longrightarrow$ LTI Memoryless $\Updownarrow$, $\alpha \in \mathbb{C}$

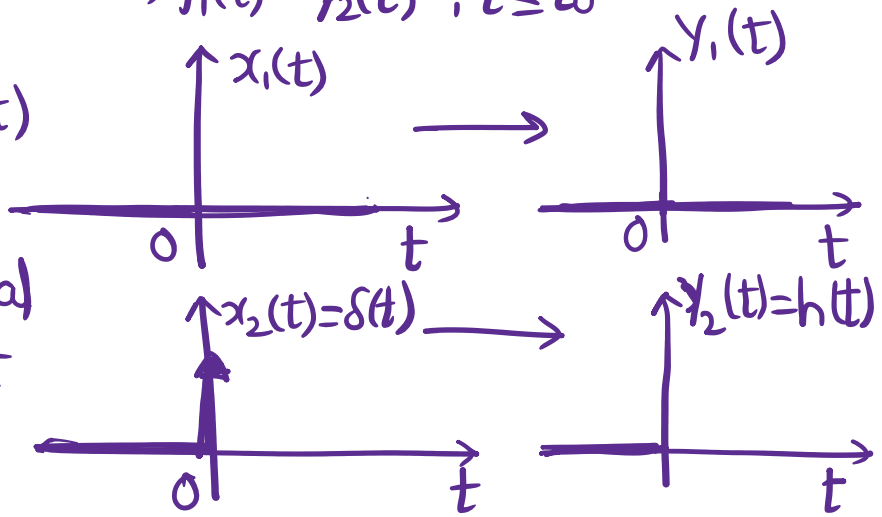
$h(t) = \alpha \delta(t) \longrightarrow$ Impulse response

$\therefore$ Memoryless LTI $\Longleftrightarrow h(t) = \alpha \delta(t)$

2) Causal/NC - LTI:



$x_1(t) = x_2(t)$, $t \le t_0$
$\Rightarrow y_1(t) = y_2(t)$, $t \le t_0$



$$\left.\begin{aligned} &x_1(t) = x_2(t),\ t<0 \\ \Rightarrow\ &y_1(t) = y_2(t),\ t<0 \\ \Rightarrow\ &0 = h(t),\ t<0 \end{aligned}\right\} \text{Causal LTI}$$

Causal LTI $\Rightarrow h(t) = 0$, $t<0$

$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau \longrightarrow$ LTI

Present/past values of i/p

& $h(t) = 0$, $t<0 \Rightarrow y(t) = \int_{0}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$

$\Rightarrow$ Causal

$\therefore \quad h(t)=0,\ t<0 \Leftrightarrow$ Causal LTI

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3) <u>Stable/Unstable LTI</u>:

$$y(t)=\int_{-\infty}^{\infty} h(\tau)x(t-\tau)\,d\tau$$

Let $x(t)$ be bounded $\Rightarrow |x(t)|\le B_x<\infty \quad \forall t$

$$\Rightarrow |y(t)| \le \left|\int_{-\infty}^{\infty} h(\tau)x(t-\tau)\,d\tau\right| \le \int_{-\infty}^{\infty} |h(\tau)x(t-\tau)|\,d\tau$$

$$\le B_x \underbrace{\int_{-\infty}^{\infty} |h(\tau)|\,d\tau}_{<\infty}$$

$$\underline{\underline{B_y<\infty}}$$

$\therefore \int_{-\infty}^{\infty} |h(t)|\,dt<\infty \Rightarrow$ Stable LTI

Let system be stable <u>LTI</u>

$$\Rightarrow y(t)=\int_{-\infty}^{\infty} h(\tau)x(t-\tau)\,d\tau$$

$$\Rightarrow y(0)=\int_{-\infty}^{\infty} h(\tau)x(-\tau)\,d\tau$$

Let $x(-t) \triangleq \operatorname{sgn}(h(t)) = \begin{cases} 1, & h(t)>0 \\ -1, & h(t)\le 0 \end{cases} \Big\}\ |x(t)|=1$

$\Rightarrow$ <u>i/p $x(t)$ is bounded</u>.

$$\& \ y(0)=\int_{-\infty}^{\infty} h(\tau)\operatorname{sgn}(h(\tau)) = \int_{-\infty}^{\infty} |h(\tau)|\,d\tau < \infty$$

$\therefore$ Stable, LTI $\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|\,dt<\infty$